MIL-P-18716B(OS)
7 February, 1977
Superseding
MIL-P-18716A(OS)
17 December 1971

MILITARY SPECIFICATION

PRIMER, ELECTRIC, MARK 48 MOD 2

This specification is approved for use by the Naval Sea Systems Command (OS), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the minimum requirements for the fabrication, preparation for delivery, and acceptance of the Electric Primer, Mark 48 Mod 2, critical item, hereinafter referred to as the primer (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

LLL-B-631	Boxes, Fiber Corrugated

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	Inspection, Radiographic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head Maryland 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1390

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DRAWINGS

Naval Sea Systems Command (Code Ident 10001)

DL 2434756	Primer Electric Mark 48 Mod 2
LD 296045	Primer Container Mark 11 Mod 6 (Inner)
2847234	Charge, Propelling 5-Inch, 38-Caliber Full, Mark 63 Mod 0
512849	Primer Containers Mark 7 Mods 0 & 1 Outer for Primers

PUBLICATIONS

Naval Sea Systems Command (Code Ident 10001)

WR-43A	Preparation of Quality Assurance Provisions
WR-53/819	Palletizing Domestic Unit Load
WR-53/851	Palletizing Domestic Unit Load
OD 49540	Functioning Test of Electric Primers

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49 CFR 100-199	Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20462. Orders should specify "49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 First article. Unless otherwise specified in the contract or order (see 6.2.1), a first article shall be manufactured, inspected, and accepted prior to production. Component inspection shall be part of that requirement for all first articles. The Government may also require, at their discretion, the manufacture, inspection, and acceptance of a first article if any of the following conditions exist:

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(a) There has been a break in production of at least six months duration;

(b) The supplier has made a major change in his assembly procedures;

(c) There has been a major change in production personnel.

First articles shall meet the manufacture procedures and inspection requirements specified herein.

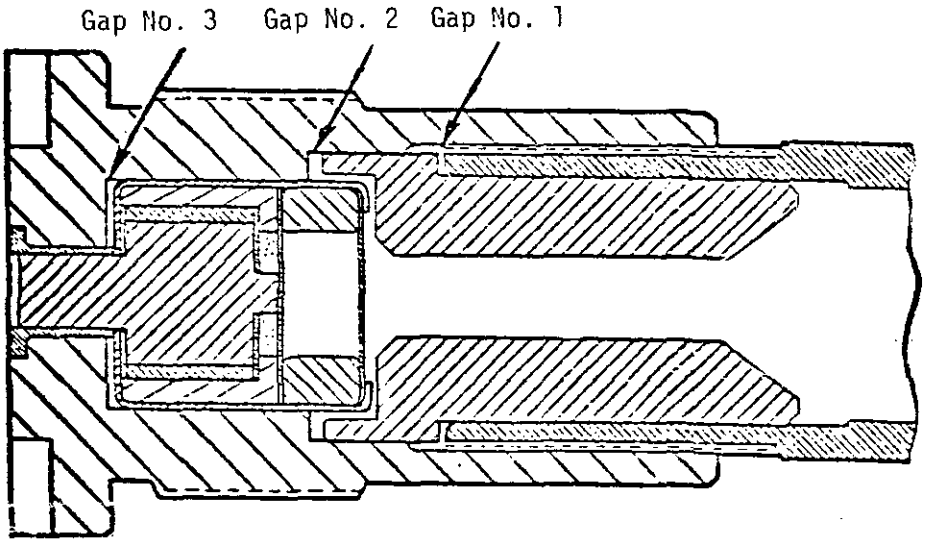
3.2 Characteristics. The primer shall conform to all the requirements specified herein and specified in all the documents listed on DL 2434756.

3.2.1 Radiographic. (M101) The primer shall not exhibit any discernible gaps as identified in Figure 1.

3.2.2 Detail strip. The primer and all its components shall conform to the following requirements:

(a) (C1) Each critical characteristic shall be as specified in the applicable documents listed on DL 2434756.

(b) (M102) Each major characteristic shall be as specified in the applicable documents listed on DL 2434756.



GAP IDENTIFICATION

FIGURE 1

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3.2.3 Resistance. (M103) The primer shall have a resistance of .10 to .18 ohms.

3.2.4 Functioning. The primer shall meet the following criteria:

(a) (M104) Primer functioning time shall have an upper specification limit of .006 seconds;

(b) (M105) The primer shall not misfire (see 6.5.1);

(c) (M106) The primer shall not exhibit blowback (see 6.5.2); and

(d) (C2) The primer shall not exhibit a split stock, split tube or detachment of metallic components other than blowout of ignition element closing discs.

3.2.5 Vibration. The primer shall meet the following requirements:

(a) (C3) The primer shall not function during vibration;

(b) (C4) The stock and tube shall maintain material integrity, and the primer shall maintain assembly integrity at the mating threads of the stock and tube;

(c) (M107) The primer shall not otherwise physically breakdown during vibration;

(d) The primer shall meet the requirements of 3.2.3 after vibration; and

(e) The primer shall meet the requirements of 3.2.4 after vibration.

3.2.6 Ballistic. The primer shall provide ignition which will meet the following criteria:

(a) (C5) The primer shall not exhibit a split stock, split tube or detachment of metallic components other than blowout of ignition element closing discs;

(b) (M108) The primer shall not misfire;

(c) (M109) The ejection time of the projectile from the gun barrel shall not be greater than .050 seconds after application of firing energy;

(d) (M110) There shall be no blowback; and

(e) (M111) The initial velocity standard deviation shall not be greater than sixteen feet per second.

3.3. Workmanship. All parts shall be free of chips, dirt, grease, rust, and all other foreign materials. There shall be no burrs or other imperfections of manufacture which would affect the assembly, functioning, or handling of the product. The cleaning methods used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First Article Inspection (see 4.3)

(b) Quality Conformance Inspection (see 4.4)

4.3 First article. The intent of the first article is to demonstrate that the contractor's fabrication methods are capable of yielding primers that comply with the technical requirements of the contract or purchase order.

4.3.1 Formation. The first article shall be a homogeneous sample of 224 assembled primers and five each of every component part and subassembly (less explosive components) fabricated, packaged, and packed in accordance with the requirements of this specification and the contract or purchase order using the methods and procedures to be used for production.

4.3.2 Inspection. The first article shall be submitted to the facility designated in the contract or order (see 6.2.1) for test and examination in accordance with the First Article Inspection Sampling Plan of Table 1.

4.4 Production

4.4.1 Lot formation. Unless otherwise specified in the contract or purchase order (see 6.2.1), production lots shall consist of not more than 20,000 homogeneous primers plus those required for inspection (see 4.4.2). Each production lot shall contain only one class 2 black powder lot and one class 4 black powder lot and only one tubing heat treat lot.

4.4.2 Inspection. Unless otherwise specified in the contract or purchase order, 168 samples shall be selected at random from each production lot for test and examination in accordance with the Quality Conformance Inspection Sampling Plan of Table 1.

4.5 Classification of characteristics. The characteristics verified by the tests and examination herein are classified as critical, major, or minor in accordance with WR-43A. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

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TABLE 1

FIRST ARTICLE AND PRODUCTION INSPECTION SAMPLING PLANS

TEST OR EXAMINATION	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	FIRST ARTICLE INSPECTION SAMPLE SIZE	FIRST ARTICLE INSPECTION ACCEPTANCE CRITERIA ACCEPT	FIRST ARTICLE INSPECTION ACCEPTANCE CRITERIA REJECT	QUALITY CONFORMANCE INSPECTION SAMPLE SIZE	QUALITY CONFORMANCE INSPECTION ACCEPTANCE CRITERIA ACCEPT	QUALITY CONFORMANCE INSPECTION ACCEPTANCE CRITERIA REJECT
RADIOGRAPHIC (M101)	3.2.1	4.6.1	224[1]	0	1	168[1]	0	1
DETAIL STRIP	3.2.2	4.6.2						
a. (C1)			56	0	1	56	0	1
b. (M102)			30[2]	1	2	30[2]	1	2
RESISTANCE (M103)	3.2.3	4.6.3	224[1]	2	3	168[1]	2	3
FUNCTIONING								
a. (M104)	3.2.4	4.6.4	MIL-STD-414	AQL ≤.4%	AQL >.4%	MIL-STD-414	AQL ≤.4%	AQL >.4%
b. (M105) (M106)			SAMPLE SIZE	0	1	SAMPLE SIZE	0	1
c. (C2)			CODE LTR. O	0	1	CODE LTR. N	0	1
VIBRATION								
a. (C3) (C4)	3.2.5	4.6.5	MIL-STD-414	0	1	MIL-STD-414	0	1
b. (M107)			SAMPLE SIZE	0	1	SAMPLE SIZE	0	1
c. (M103)	(3.2.3)	(4.6.3)	CODE LTR. J	0	1	CODE LTR. H	0	1
d. (M104)	(3.2.4)	(4.6.4)		AQL ≤.4%	AQL >.4%		AQL ≤.4%	AQL >.4%
e. (M105) (M106) (C2)				0	1		0	1
BALLISTIC	3.2.6	4.6.6						
a. (C5)			20	0	1			
b. (M108) (M109) (M110)				0	1			
c. (M111)				a ≤16	a >16			
PACKAGING								
a.	5.1							
b. (M112)	5.1.1	4.6.7	5[3]	0	1	3[3]	0	1
PACKING	5.2	4.6.7	5[4]	0	1	3[4]	0	1
MARKING (M113)	5.3	4.6.7	5[3,4]	0	1	3[3,4]	0	1

[1] Radiographic examination and resistance test shall be performed on all samples prior to any other test or examination.
[2] Sample as same as for (a) above. Acceptance criteria is based on the number of defects for each major characteristic.
[3] Sample size of MK 11 containers
[4] Sample size of MK 7 containers

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4.6 Test and examination methods (see 6.4). Unless otherwise specified in the contract or purchase order, the contractor and all test activities identified in the contract or purchase order shall submit to the technical activity and design agent for review, a description of the proposed procedures and equipment to be used to perform all examinations and tests. Primers shall be selected at random for each test or examination. Identification of each primer's complete test and examination results shall be maintained. Any primer (or part thereof) exhibiting a defect shall be secured for additional examination until otherwise directed by the procuring activity. A defective is any primer which fails to meet the associated requirements for the examination and tests described in Table 1 (paragraph 4.3.2).

4.6.1 Radiographic. The primer shall be subjected to a radiographic inspection in accordance with MIL-STD-453. The primer shall be mounted in the vertical position with the primer stock up and X-rayed perpendicular to its axis, focused at a point 0.6 ± 0.2 inch from the stock face. Any primer which meets the requirements of 3.2.1 but appears to have been misassembled shall be subjected to detail strip and shall meet the requirements of 3.2.2.

4.6.2 Detail strip. The primer shall be disassembled to determine conformance to all the discernible critical and major characteristics specified on the applicable documents listed on DL 2434756. All inspections shall be performed at the highest level of assembly possible.

4.6.3 Resistance. The primer resistance shall be measured by using a test circuit which limits the bridge current to 50 milliamperes maximum. The primer stock and the electrode shall be used as terminals. Fifteen to twenty pounds force shall be applied to the electrode during resistance measurement with a probe having a .050 ± .005 spherical radius. Assure that contact on clean bright surfaces is obtained.

4.6.4 Functioning. The primer shall be subjected to the functioning test of OD 49540. Primer functioning time shall be evaluated in accordance with MIL-STD-414, Section B.

4.6.5 Vibration. The primer shall be subjected to the transportation vibration test of MIL-STD-331, Test 104, Procedure II, except that the orientation of each primer during this three phase test shall be as follows:

(a) Phase I: Primer axis horizontal;

(b) Phase II: Primer axis vertical; primer stock down; and

(c) Phase III: Primer axis vertical, primer stock up.

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The primer shall be marked with a line in the axial direction at the junction of the stock and tube indexing their initial relative position, in order to determine the integrity after vibration. The primer shall be rigidly mounted to the vibration table at the primer stock. After vibration, the primer shall be subjected to the tests of 4.6.3 and 4.6.4.

4.6.6 Ballistic. The primer shall be assembled into a 5"/38 caliber full service propelling charge assembly in accordnance with Drawing 2847234. The propellant shall be of a master lot as defined in 6.5.3 and shall be conditioned at 90 ± 15 degrees Fahrenheit for 24 hours minimum prior to assembly of the propelling charge. All other components shall be of serviceable lots and shall be conditioned at 85 ± 20 degrees Fahrenheit for 24 hours minimum prior to assembly of the propelling charge. The assembled propelling charges shall be conditioned at 90 ± 5 degrees Fahrenheit for 48 hours minimum immediately prior to firing. Serviceable blind, loaded, and plugged projectiles shall be used. The rounds shall be fired in a 5"/38 electrically fired gun with a first quarter life barrel. Initial projectile muzzle velocity and ejection time shall be recorded. Initial velocity standard deviation, s, shall be calculated as follows:

$$s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$$, feet per second; where,

x = velocity of each firing, feet per second;
n = number of firings; and
$\bar{x}$ = mean velocity = $\frac{\Sigma x}{n}$, feet per second.

4.6.7 Packaging, packing, and marking. The packaging, packing, and marking shall be visually examined for conformance to the major requirements of Section 5.

5. PACKAGING (see 4.6.7)

5.1 Packaging (Level A). The primer shall be packaged in accordance with applicable drawing listed on LD 296045.

5.1.1 Inert Components. (M112) Components and subassemblies shall be packaged in commerical fiberboard containers meeting the requirements of Federal Specification LLL-B-631. Each container shall incorporate corrugated or solid fiberboard dividers or collars suitable for preventing damage in handling and transportation. The gross weight of the container contents shall not be greater than 70 pounds.

5.2 Packing (Level A). Primers packaged in accordance with 5.1 shall be packed in accordance with Drawing 512849.

5.3 Palletization. Palletization of primers packed in accordance with 5.2 shall be in accordance with WR-53/819 or WR-53/851 as applicable.

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5.4 Marking. (M113) In addition to any special marking required by the contract or order, all markings shall be in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-179.

5.5 Data cards. As specified in the contract or order (see 6.2), one data card shall be placed in the top of each pack (see 5.2), five shall be forwarded to the procuring activity, and twenty shall go to the ammunition depot receiving the primers. For Navy procurements, two data cards shall also go to each of the activities listed in 6.4.

6. NOTES

6.1 Intended use. The Electric Primer MK 48 MOD 2 is intended for use as the initiator in the MK 63 MODS 5"/38 caliber full propelling charges, in the MK 64 MODS 5"/38 caliber reduced propelling charges, in the MK 65 MOD 0 5" common clearing charge.

6.2 Ordering data. DD Form 1423 and other related procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) Whether a first article sample is required (see 3.1).

(c) The facility designated to perform first article inspection (see 4.3.2).

(d) Production lot quantity (ies) if other than specified in 4.4.1.

(e) Production inspection sample quantity if other than maximum specified in 4.4.2.

(f) The facility (ies) designated to perform production inspection if other than the contractor (see 4.1 and 4.4.2).

(g) Safety precautions (see 6.3).

6.2.2 Contract data requirements. The item of deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
5.5	Ammunition Data Card	DI-E-2001

DID's (Data Item Descriptions/DD Forms 1664) for the above data requirements are documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DID's when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

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6.3 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Armed Services Procurement Regulations (ASPR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.4 Test procedures and reporting. For Navy procurements, the procuring activity should include in each work authorization for first article and quality conformance inspections, the requirement for the preparation and forwarding to the activities listed below of all test and examination procedures prior to performance of any inspection. The work authorization should also include the requirement for the forwarding of all test data in an intelligible form to the activities listed below within five working days of the completion of the tests.

Commander
Naval Sea Systems Command
NAPEC (SEA-99262)
Naval Weapons Support Center
Crane, Indiana 47522

Command Officer
Naval Ordnance Station
(Code 5231)
Indian Head, Maryland 20640

6.5 Definitions

6.5.1 Misfire. Misfire is the failure of the main black powder charge of the primer to ignite when the primer is subjected to the appropriate firing source.

6.5.2 Blowback. Blowback is gas leakage around or through the stock end of the primer and is verified by observance during firing or by visual evidence of residue after firing.

6.5.3 Master powder. A master powder is a lot of powder specifically so designated by the Naval Sea Systems Command. It has well established ballistics near the center of the ballistics specification range for powders in its category. There is at least one master powder for different nominal propelling compositions within a gun caliber. The master powder has an assigned service charge and an assigned standard pressure obtained either from a number of new gun firings or from matching a previous master powder.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Preparing Activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-P-18716B(OS) Primer, Electric, Mark 48 Mod 2

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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